

Nikolaos Stathoulopoulos

PH.D. CANDIDATE IN ROBOTICS & AI · CO-FOUNDER, SAPHAI AB

Gothenburg, Västra Götaland, Sweden

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“Towards Lifelong Autonomy for Heterogeneous Multi-Robot Systems”

Summary

A Ph.D. candidate in the Robotics & AI (RAI) Group at Luleå University of Technology, Sweden, working at the intersection of localization, mapping, and 3D perception. My research centers on multi-robot and multi-session SLAM and long-term map management, combining classical and learning-based methods to build adaptive, efficient, and reliable perception systems. Core topics include heterogeneous place recognition, loop closures, and back-end optimization for robust lifelong operation. Co-founder of SAPHAI AB, and an active member of the IEEE Robotics & Automation Society (RAS) and the broader robotics community, with an emphasis on open-source, reproducible research.

Highlights

- Published across established venues in robotics (IEEE ICRA, IEEE/RSJ IROS, IEEE RA-L) and field robotics (IEEE T-FR, JFR).
- Release open-source code for my research and teaching, prioritising *reproducibility and reuse*:
 - e.g., author of hello-slam [📄 ★ 264], the SLAM teaching material used in the Advanced Robotics course at LTU, plus FRAME [📄 ★ 69], opt-key [📄 ★ 36], sga-dpcc [📄 ★ 24], and InLiER [📄 ★ 9]
- Academic honors & awards:
 - IVA's 100 List 2024 — current research with potential to create value, Royal Swedish Academy of Engineering Sciences
 - Best Paper Award at the *Standing the Test of Time* Workshop, IEEE/RSJ IROS 2024
- Research & entrepreneurial experience:
 - Co-founder of SAPHAI AB, building on research in AI-enabled data-driven change detection
 - Awarded Wallenberg Launchpad (WALP) Phase 2 after a successful pitch
 - Contributor to five EU Horizon and national projects on autonomous mining and construction robotics
 - Industry collaboration with Ericsson on 5G-edge-assisted robotics and communication-aware map sharing
- Teaching & service:
 - Teaching assistant for five courses at LTU and supervisor of three M.Sc. thesis students
 - Reviewer for IEEE RA-L, ICRA, IROS, and Robotics and Autonomous Systems since 2024

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- SLAM using range-based sensors (LiDAR, radar)
- Multi-robot and multi-session SLAM
- Long-term map management and lifelong autonomy
- LiDAR-based place recognition, including heterogeneous sensor setups
- Loop closure detection and back-end pose graph optimization
- Keyframe sampling and map sparsification for large-scale SLAM
- Point cloud compression and bandwidth-aware map sharing
- 3D scene graphs and scene-aware representations
- Change detection and static map building
- Field robotics for mining, construction, and subterranean applications
- Autonomous robots, including ground, aerial, and quadruped robots
- Recovering from failures and uncertainty-aware robotic systems

Education

Luleå University of Technology (LTU)

PH.D. CANDIDATE, ROBOTICS & AI (RAI) GROUP

Luleå, Sweden

Sep. 2022 - Present

- Topic: Efficient, Scalable and Reliable Multi-Robot/Session SLAM for Lifelong Autonomy
- Supervised by Prof. George Nikolakopoulos, Robotics & AI Group

University of Patras (UPatras)

M.S. IN ELECTRICAL AND COMPUTER ENGINEERING (ECE)

Patras, Greece

Aug. 2020 - Aug. 2022

- Integrated master's stage of the five-year Electrical and Computer Engineering programme
- Thesis: Decentralized exploration and map building with multiple autonomous UAVs [pdf]
- Specialization: Signal, Systems and Control

University of Patras (UPatras)

B.S. IN ELECTRICAL AND COMPUTER ENGINEERING (ECE)

Patras, Greece

Oct. 2017 - Aug. 2020

Work Experience

Co-Founder

SAPHAI AB

Sweden

Oct. 2023 - Current

- Co-founded the company around AI-enabled, data-driven change detection for industrial environments.
- Recognised on IVA's 100 List 2024 and awarded Wallenberg Launchpad (WALP) Phase 2 funding.

Teaching Assistant

ROBOTICS & AI (RAI) GROUP, LULEÅ UNIVERSITY OF TECHNOLOGY

Luleå, Sweden

Sep. 2022 - Current

- Teaching assistant for Advanced Robotics, Computer Vision & Image Processing, Robotics, and Industrial Automation.
- Prepared and delivered lectures on SLAM, along with the accompanying hello-slam open-source material.
- Supervise M.Sc. thesis students on relocalization, map compression, and change-aware LiDAR compression.

Summer Research Intern

ROBOTICS & AI (RAI) GROUP, LULEÅ UNIVERSITY OF TECHNOLOGY

Luleå, Sweden

Jun. 2022 - Jul. 2022

- Research internship preceding the start of my Ph.D. studies in the same group.

Teaching Assistant

EISLAB MACHINE LEARNING, LULEÅ UNIVERSITY OF TECHNOLOGY

Luleå, Sweden

Mar. 2022 - Jun. 2022

- Teaching assistant for Advanced Deep Learning (D7047E).

Research Project Experience

CIRCULess — Upcycling Mineral and Timber-Based Waste from Construction & Manufacturing Process Industries

RAI Group, Luleå Univ. of Technology

SUPPORTED BY THE EUROPEAN UNION, HORIZON EUROPE

2024 - Current

- Robotic perception and mapping for circular construction and manufacturing workflows.
- Keywords: 3D perception, mapping, change detection

PERSEPHONE — Autonomous Exploration and Extraction of Deep Mineral Deposits

RAI Group, Luleå Univ. of Technology

SUPPORTED BY THE EUROPEAN UNION, HORIZON EUROPE

2023 - Current

- Multi-robot localization and mapping for autonomous exploration of deep underground mines.
- Keywords: multi-robot SLAM, subterranean robotics, map merging

SP14 — Sustainable Mining of the Future

RAI Group, Luleå Univ. of Technology

SUPPORTED BY LKAB THROUGH THE SUM ACADEMY PROGRAMME

2022 - 2024

- Perception and localization technologies supporting sustainable, automated mining operations.
- Keywords: place recognition, global re-localization, change detection

NEXGEN SIMS — Efficient, Safe, and Sustainable Mining

RAI Group, Luleå Univ. of Technology

SUPPORTED BY THE EUROPEAN UNION, HORIZON 2020

2022 - 2024

- Autonomous aerial and ground robotic systems for underground mining environments.
- Keywords: field robotics, autonomous inspection, multi-agent systems

ILLuMINEation — Bright Concepts for a Safe and Sustainable Digital Mining Future

RAI Group, Luleå Univ. of Technology

SUPPORTED BY THE EUROPEAN UNION, HORIZON 2020

2022 - 2023

- Global pose estimation and map merging for digitalised mining infrastructure.
- Keywords: global localization, point cloud registration, digital mining

Publications

Bold denotes my name. Full list also available on Google Scholar and ORCID.

INTERNATIONAL JOURNAL

1. **Nikolaos Stathoulopoulos** and George Nikolakopoulos, “InLiER: Learning-Free Heterogeneous LiDAR Place Recognition via Intermediate Mixed-Radix Structural Keypoint Tokenization,” *IEEE Robot. Automat. Lett. (RA-L)*, 2026. Accepted. To appear. [\[code\]](#) [\[arXiv\]](#)
2. **Nikolaos Stathoulopoulos**, Christoforos Kanellakis, and George Nikolakopoulos, “A Minimal Subset Approach for Informed Keyframe Sampling in Large-Scale SLAM,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 11, no. 1, pp. 738–745, 2026. **Presented at IEEE ICRA’26**. [\[code\]](#) [\[arXiv\]](#) [\[video\]](#)
3. **Nikolaos Stathoulopoulos**, Christoforos Kanellakis, and George Nikolakopoulos, “Have We Scene It All? Scene Graph-Aware Deep Point Cloud Compression,” *IEEE Robot. Automat. Lett. (RA-L)*, vol. 10, no. 12, pp. 12477–12484, 2025. **Presented at IEEE ICRA’26**. [\[code\]](#) [\[arXiv\]](#) [\[video\]](#)
4. Niklas Dahlquist, Samuel Nordström, **Nikolaos Stathoulopoulos**, Björn Lindqvist, Akshit Saradagi, and George Nikolakopoulos, “Deployment of an Aerial Multi-agent System for Automated Task Execution in Large-scale Underground Mining Environments,” *IEEE Trans. Field Robot. (T-FR)*, vol. 2, pp. 997–1101, 2025. **Presented at IEEE ICRA’26**. [\[arXiv\]](#) [\[video\]](#)
5. Samuel Nordström, **Nikolaos Stathoulopoulos**, Niklas Dahlquist, Björn Lindqvist, Ilias Tevetzidis, Christoforos Kanellakis, and George Nikolakopoulos, “Safety Inspections and Gas Monitoring in Hazardous Mining Areas After Blasting Using Autonomous UAVs,” *J. Field Robot. (JFR)*, vol. 42, pp. 2076–2094, 2025. [\[link\]](#) [\[video\]](#)
6. **Nikolaos Stathoulopoulos**, Björn Lindqvist, Anton Koval, Ali-Akbar Agha-Mohammadi, and George Nikolakopoulos, “FRAME: A Modular Framework for Autonomous Map Merging: Advancements in the Field,” *IEEE Trans. Field Robot. (T-FR)*, vol. 1, pp. 1–26, 2024. **Presented at IEEE ICRA’25**. [\[code\]](#) [\[arXiv\]](#)
7. Gerasimos Damigos, **Nikolaos Stathoulopoulos**, Anton Koval, Tore Lindgren, and George Nikolakopoulos, “Communication-Aware Control of Large Data Transmissions via Centralized Cognition and 5G Networks for Multi-Robot Map Merging,” *J. Intell. Robot. Syst. (JIRS)*, vol. 110, art. no. 22, 2024. [\[link\]](#)
8. **Nikolaos Stathoulopoulos**, Anton Koval, and George Nikolakopoulos, “3DEG: Data-driven Descriptor Extraction for Global Re-localization in Subterranean Environments,” *Expert Syst. Appl. (ESWA)*, vol. 237(B), art. no. 121508, 2024. [\[link\]](#) [\[arXiv\]](#)
9. **Nikolaos Stathoulopoulos**, Anton Koval, and George Nikolakopoulos, “A Comparative Field Study of Global Pose Estimation Algorithms in Subterranean Environments,” *Int. J. Control, Automat. Syst. (IJCAS)*, vol. 22, pp. 690–704, 2024. [\[link\]](#)

INTERNATIONAL CONFERENCE

1. Gerasimos Damigos, **Nikolaos Stathoulopoulos**, Achilleas Santi Seisa, Sara Sandberg, and George Nikolakopoulos, “Real-time Point Cloud Data Transmission via L4S for 5G-Edge-Assisted Robotics,” in *Proc. IFAC World Congress*, Busan, South Korea, 2026. Accepted. To appear. [\[code\]](#) [\[arXiv\]](#)
2. Akash Patel, Mario Alberto Valdes Saucedo, **Nikolaos Stathoulopoulos**, Viswa Narayanan Sankaranarayanan, Ilias Tevetzidis, Christoforos Kanellakis, and George Nikolakopoulos, “A Hierarchical Graph-Based Terrain-Aware Autonomous Navigation Approach for Complementary Multimodal Ground-Aerial Exploration,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Atlanta, GA, USA, 2025, pp. 15879–15885. [\[arXiv\]](#)
3. Mario Alberto Valdes Saucedo, **Nikolaos Stathoulopoulos**, Akash Patel, Christoforos Kanellakis, and George Nikolakopoulos, “Leveraging Computation of Expectation Models for Commonsense Affordance Estimation on 3D Scene Graphs,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Abu Dhabi, UAE, 2024, pp. 9797–9802. [\[arXiv\]](#) [\[video\]](#)
4. **Nikolaos Stathoulopoulos**, Mario Alberto Valdes Saucedo, Anton Koval, and George Nikolakopoulos, “RecNet: An Invertible Point Cloud Encoding through Range Image Embeddings for Multi-Robot Map Sharing and Reconstruction,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Yokohama, Japan, 2024, pp. 4883–4889. [\[arXiv\]](#) [\[video\]](#)
5. **Nikolaos Stathoulopoulos**, Emanuele Pagliari, Luca Davoli, and George Nikolakopoulos, “Redundant and Loosely Coupled LiDAR-Wi-Fi Integration for Robust Global Localization in Autonomous Mobile Robotics,” in *Proc. Int. Conf. Adv. Robot. (ICAR)*, Abu Dhabi, UAE, 2023, pp. 121–127. [\[arXiv\]](#)

6. **Nikolaos Stathoulopoulos**, Anton Koval, and George Nikolakopoulos, “Irregular Change Detection in Sparse Bi-Temporal Point Clouds Using Learned Place Recognition Descriptors and Point-to-Voxel Comparison,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Detroit, MI, USA, 2023, pp. 8077–8083. [\[arXiv\]](#)
7. **Nikolaos Stathoulopoulos**, Anton Koval, Ali-Akbar Agha-Mohammadi, and George Nikolakopoulos, “FRAME: Fast and Robust Autonomous 3D Point Cloud Map-Merging for Egocentric Multi-Robot Exploration,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, London, UK, 2023, pp. 3483–3489. [\[code\]](#) [\[arXiv\]](#)
8. Dimitris Dimitropoulos, **Nikolaos Stathoulopoulos**, Eleni Rompola, et al., “School Hackerspaces: Supporting Informatics in Secondary Education,” in *Proc. Conf. Informatics Educ. (CIE)*, 2020, pp. 88–95. [\[pdf\]](#)

WORKSHOP

1. **Nikolaos Stathoulopoulos**, Vidya Sumathy, Christoforos Kanellakis, and George Nikolakopoulos, “Does Sample Space Matter? Preliminary Results on Keyframe Sampling Optimization for LiDAR-based Place Recognition,” in *Standing the Test of Time Workshop: Retrospective and Future of World Representations for Lifelong Robotics*, at *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Abu Dhabi, UAE, 2024. **Received Best Paper Award.** [\[code\]](#) [\[pdf\]](#)

Teaching Experience

COURSES

[R7021E] Advanced Robotics

TEACHING ASSISTANT & GUEST LECTURER ON SLAM

- Delivered three lectures on SLAM, with accompanying open-source material: [hello-slam](#)

Luleå University of Technology

2025, 2026

[R7020E] Computer Vision & Image Processing

TEACHING ASSISTANT

Luleå University of Technology

2024, 2025

[R7010E] Robotics

TEACHING ASSISTANT

Luleå University of Technology

2023

[R7008E] Industrial Automation

TEACHING ASSISTANT

Luleå University of Technology

2023

[D7047E] Advanced Deep Learning

TEACHING ASSISTANT

Luleå University of Technology

2022

M.SC. THESIS SUPERVISION

Optimizing Relocalization Likelihood in Challenging Environments With Reinforcement Learning

LOUISE NILSSON

Luleå University of Technology

2025

Task-Driven Map Compression for Collaborative Exploration of Ground-Aerial Robot Teams

HAMPUS KÄMPPI

Luleå University of Technology

2025

Adaptive Point Cloud Compression for Resource Constrained Lifelong SLAM

MORITZ ENDEMANN

Luleå University of Technology

2025 - Present

Skills

Programming	Python, MATLAB, C++, C, $\LaTeX$
ML, Vision & Geometry	PyTorch, Open3D, OpenCV, NumPy, PCL, SciPy, GTSAM
Robotics & Workflow	ROS / ROS2, Linux, Git, Conda, Docker
Languages	Greek (native), English (OLS C2)

Honors & Awards

AWARDS

- 2024 **IVA's 100 List — Research with Potential to Create Value**, Royal Swedish Academy of Engineering Sciences (IVA), for *AI-enabled Data-Driven Change Detection* Sweden
- 2024 **Best Paper Award**, Standing the Test of Time Workshop, in *IEEE/RSJ IROS'24* Abu Dhabi, UAE

GRANTS & SCHOLARSHIPS

- 2025 **RAS Travel Support Grant**, IEEE Robotics and Automation Society, for *IEEE ICRA'25* Atlanta, GA, USA
- 2024 **Jubilee Grant**, Knut and Alice Wallenberg Foundation, for *IEEE ICRA'24* Yokohama, Japan
- 2023 **Wallenberg Launchpad (WALP) Phase 2**, WASP / Knut and Alice Wallenberg Foundation, for *AI-enabled Data-Driven Change Detection* Sweden

Academic Services

- 2026 **Invited Reviewer, Special Session**, IEEE ETFA
- 2026 **Reviewer**, IEEE RA-L, IEEE ICRA, IEEE/RSJ IROS
- 2025 **Reviewer**, IEEE RA-L, IEEE ICRA, IEEE/RSJ IROS, Robotics and Autonomous Systems
- 2024 **Reviewer**, IEEE RA-L, IEEE ICRA, Springer DARS